

# full\_robustness\_sensitivity\_results

| Family           | Gate  | Scenario               | Specification | N   | Term         | Metric | Estimate    | CI_Low     | CI_High    | p_value     | Significant | Direction | Formatted               |
|------------------|-------|------------------------|---------------|-----|--------------|--------|-------------|------------|------------|-------------|-------------|-----------|-------------------------|
| Repository-level | Gate1 | Repository-clustered   | C/SE + PR     | 312 | Context      | OR     | 2.14309     | 1.0902     | 4.21285    | 0.0270784   | True        | positive  | 2.14* [1.09, 4.21]      |
| Repository-level | Gate1 | Repository-clustered   | C/SE + PR     | 312 | Specificity  | OR     | 65.8384     | 9.23054    | 469.604    | 2.95198e-05 | True        | positive  | 65.84*** [9.23, 469.60] |
| Repository-level | Gate1 | Repository-clustered   | C/SE + PR     | 312 | Verification | OR     | 0.903855    | 0.418553   | 1.95185    | 0.796908    | False       | negative  | 0.90 [0.42, 1.95]       |
| Repository-level | Gate1 | Repository-clustered   | C/SE + PR     | 312 | Log_PR_Size  | Size   | 1.11543     | 0.906835   | 1.37202    | 0.301054    | False       | positive  | 1.12 [0.91, 1.37]       |
| Repository-level | Gate1 | Repository-clustered   | C/SE + PR     | 142 | Context      | OR     | 2.21887     | 1.00231    | 4.91201    | 0.0493378   | True        | positive  | 2.22* [1.00, 4.91]      |
| Repository-level | Gate1 | Repository-clustered   | C/SE + PR     | 142 | Specificity  | OR     | 0.792625    | 0.300109   | 2.09342    | 0.639061    | False       | negative  | 0.79 [0.30, 2.09]       |
| Repository-level | Gate1 | Repository-clustered   | C/SE + PR     | 142 | Verification | OR     | 8.45432     | 3.51858    | 20.3138    | 1.81723e-06 | True        | positive  | 8.45*** [3.52, 20.31]   |
| Repository-level | Gate1 | Repository-clustered   | C/SE + PR     | 142 | Log_PR_Size  | Size   | 1.36131     | 1.0669     | 1.73695    | 0.0131089   | True        | positive  | 1.36* [1.07, 1.74]      |
| Repository-level | Gate2 | Repository-clustered   | C/SE + PR     | 892 | Context      | AME    | 0.12891     | 0.0213987  | 0.236422   | 0.01877     | True        | positive  | 0.129* [0.021, 0.236]   |
| Repository-level | Gate2 | Repository-clustered   | C/SE + PR     | 892 | Specificity  | AME    | -0.0138781  | -0.135158  | 0.107401   | 0.822539    | False       | negative  | -0.014 [-0.135, 0.107]  |
| Repository-level | Gate2 | Repository-clustered   | C/SE + PR     | 892 | Verification | AME    | -0.00846394 | -0.109619  | 0.0926906  | 0.869734    | False       | negative  | -0.008 [-0.110, 0.093]  |
| Repository-level | Gate2 | Repository-clustered   | C/SE + PR     | 892 | Log_PR_Size  | AME    | -0.0371757  | -0.077963  | 0.00361157 | 0.740319    | False       | negative  | -0.037 [-0.078, 0.004]  |
| Dominant repo    | Gate1 | Stirling largest repos | C/SE + PR     | 202 | Context      | OR     | 1.98122     | 1.03873    | 3.77887    | 0.0379583   | True        | positive  | 1.98* [1.04, 3.78]      |
| Dominant repo    | Gate1 | Stirling largest repos | C/SE + PR     | 202 | Specificity  | OR     | 63.0842     | 8.52229    | 466.966    | 4.95196e-05 | True        | positive  | 63.08*** [8.52, 466.97] |
| Dominant repo    | Gate1 | Stirling largest repos | C/SE + PR     | 202 | Verification | OR     | 0.889567    | 0.423009   | 1.87071    | 0.757665    | False       | negative  | 0.89 [0.42, 1.87]       |
| Dominant repo    | Gate1 | Stirling largest repos | C/SE + PR     | 202 | Log_PR_Size  | Size   | 1.09937     | 0.899223   | 1.34406    | 0.355505    | False       | positive  | 1.10 [0.90, 1.34]       |
| Dominant repo    | Gate1 | Stirling top 5 repos   | C/SE + PR     | 182 | Context      | OR     | 1.91956     | 0.984598   | 3.74234    | 0.0555697   | True        | positive  | 1.92 [0.98, 3.74]       |
| Dominant repo    | Gate1 | Stirling top 5 repos   | C/SE + PR     | 182 | Specificity  | OR     | 53.2761     | 7.14593    | 397.198    | 0.000105077 | True        | positive  | 53.28*** [7.15, 397.20] |
| Dominant repo    | Gate1 | Stirling top 5 repos   | C/SE + PR     | 182 | Verification | OR     | 0.907773    | 0.415885   | 1.98144    | 0.808039    | False       | negative  | 0.91 [0.42, 1.98]       |
| Dominant repo    | Gate1 | Stirling top 5 repos   | C/SE + PR     | 182 | Log_PR_Size  | Size   | 1.09962     | 0.890221   | 1.35828    | 0.378272    | False       | positive  | 1.10 [0.89, 1.36]       |
| Dominant repo    | Gate1 | Stirling largest repos | C/SE + PR     | 122 | Context      | OR     | 2.18963     | 0.945054   | 5.07325    | 0.0675284   | True        | positive  | 2.19 [0.95, 5.07]       |
| Dominant repo    | Gate1 | Stirling largest repos | C/SE + PR     | 122 | Specificity  | OR     | 0.789568    | 0.289809   | 2.15113    | 0.644057    | False       | negative  | 0.79 [0.29, 2.15]       |
| Dominant repo    | Gate1 | Stirling largest repos | C/SE + PR     | 122 | Verification | OR     | 8.58932     | 3.55443    | 20.7562    | 1.77856e-06 | True        | positive  | 8.59*** [3.55, 20.76]   |
| Dominant repo    | Gate1 | Stirling largest repos | C/SE + PR     | 122 | Log_PR_Size  | Size   | 1.32064     | 1.04756    | 1.6649     | 0.0186177   | True        | positive  | 1.32* [1.05, 1.66]      |
| Dominant repo    | Gate1 | Stirling top 5 repos   | C/SE + PR     | 122 | Context      | OR     | 2.36434     | 0.972827   | 5.74623    | 0.0575433   | True        | positive  | 2.36 [0.97, 5.75]       |
| Dominant repo    | Gate1 | Stirling top 5 repos   | C/SE + PR     | 122 | Specificity  | OR     | 0.755377    | 0.264233   | 2.15943    | 0.600646    | False       | negative  | 0.76 [0.26, 2.16]       |
| Dominant repo    | Gate1 | Stirling top 5 repos   | C/SE + PR     | 122 | Verification | OR     | 7.69126     | 3.03216    | 19.5094    | 1.74135e-05 | True        | positive  | 7.69*** [3.03, 19.51]   |
| Dominant repo    | Gate1 | Stirling top 5 repos   | C/SE + PR     | 122 | Log_PR_Size  | Size   | 1.36428     | 1.06831    | 1.74225    | 0.0127891   | True        | positive  | 1.36* [1.07, 1.74]      |
| Dominant repo    | Gate2 | Stirling largest repos | C/SE + PR     | 82  | Context      | AME    | 0.13374     | -0.103734  | 0.371214   | 0.269678    | False       | positive  | 0.134 [-0.104, 0.371]   |
| Dominant repo    | Gate2 | Stirling largest repos | C/SE + PR     | 82  | Specificity  | AME    | -0.00175614 | -0.242457  | 0.238945   | 0.988591    | False       | negative  | -0.002 [-0.242, 0.239]  |
| Dominant repo    | Gate2 | Stirling largest repos | C/SE + PR     | 82  | Verification | AME    | -0.0162660  | -0.236384  | 0.203852   | 0.884841    | False       | negative  | -0.016 [-0.236, 0.204]  |
| Dominant repo    | Gate2 | Stirling largest repos | C/SE + PR     | 82  | Log_PR_Size  | AME    | -0.0348446  | -0.0985387 | 0.0288495  | 0.283621    | False       | negative  | -0.035 [-0.099, 0.029]  |
| Dominant repo    | Gate2 | Stirling top 5 repos   | C/SE + PR     | 72  | Context      | AME    | 0.0980935   | 0.153001   | 0.349188   | 0.443862    | False       | positive  | 0.098 [-0.153, 0.349]   |
| Dominant repo    | Gate2 | Stirling top 5 repos   | C/SE + PR     | 72  | Specificity  | AME    | 0.0528846   | 0.20172    | 0.307489   | 0.683928    | False       | positive  | 0.053 [-0.202, 0.307]   |
| Dominant repo    | Gate2 | Stirling top 5 repos   | C/SE + PR     | 72  | Verification | AME    | -0.0648588  | -0.303635  | 0.173918   | 0.59446     | False       | negative  | -0.065 [-0.304, 0.174]  |
| Dominant repo    | Gate2 | Stirling top 5 repos   | C/SE + PR     | 72  | Log_PR_Size  | AME    | -0.0398520  | -0.108899  | 0.0291946  | 0.257953    | False       | negative  | -0.040 [-0.109, 0.029]  |

| Family               | Gate  | Scenario                  | Specification    | N   | Term         | Metric | Estimate    | CI_Low     | CI_High   | p_value     | Significant | ODD | Direction | Formatted              |
|----------------------|-------|---------------------------|------------------|-----|--------------|--------|-------------|------------|-----------|-------------|-------------|-----|-----------|------------------------|
| Language sensitivity | Gate0 | Exclude dominant language | LanguagePR 150   | 150 | Context      | OR     | 1.46592     | 0.720519   | 2.98245   | 0.291219    | False       |     | positive  | 1.47 [0.72, 2.98]      |
| Language sensitivity | Gate0 | Exclude dominant language | LanguagePR 150   | 150 | Specificity  | OR     | 49.9791     | 6.64532    | 375.89    | 0.000144    | True        |     | positive  | 49.98*** [6.65, 375.9] |
| Language sensitivity | Gate0 | Exclude dominant language | LanguagePR 150   | 150 | Verification | OR     | 0.732028    | 0.323457   | 1.65668   | 0.454124    | False       |     | negative  | 0.73 [0.32, 1.66]      |
| Language sensitivity | Gate0 | Exclude dominant language | LanguagePR 150   | 150 | Log_PR_SIR   | OR     | 1.0613      | 0.859679   | 1.3102    | 0.579962    | False       |     | positive  | 1.06 [0.86, 1.31]      |
| Language sensitivity | Gate0 | Exclude top 2 languages   | LanguagePR 120   | 120 | Context      | OR     | 1.30253     | 0.59832    | 2.83557   | 0.50547     | False       |     | positive  | 1.30 [0.60, 2.84]      |
| Language sensitivity | Gate0 | Exclude top 2 languages   | LanguagePR 120   | 120 | Specificity  | OR     | 46.3384     | 6.06066    | 354.293   | 0.000218    | True        |     | positive  | 46.34*** [6.06, 354.3] |
| Language sensitivity | Gate0 | Exclude top 2 languages   | LanguagePR 120   | 120 | Verification | OR     | 0.729302    | 0.292591   | 1.81783   | 0.49814     | False       |     | negative  | 0.73 [0.29, 1.82]      |
| Language sensitivity | Gate0 | Exclude top 2 languages   | LanguagePR 120   | 120 | Log_PR_SIR   | OR     | 1.16058     | 0.921293   | 1.46202   | 0.206193    | False       |     | positive  | 1.16 [0.92, 1.46]      |
| Language sensitivity | Gate1 | Exclude dominant language | LanguagePR 100   | 100 | Context      | OR     | 3.2722      | 1.16991    | 9.15227   | 0.023883    | True        |     | positive  | 3.27* [1.17, 9.15]     |
| Language sensitivity | Gate1 | Exclude dominant language | LanguagePR 100   | 100 | Specificity  | OR     | 0.524492    | 0.156367   | 1.75927   | 0.295972    | False       |     | negative  | 0.52 [0.16, 1.76]      |
| Language sensitivity | Gate1 | Exclude dominant language | LanguagePR 100   | 100 | Verification | OR     | 10.3296     | 3.64877    | 29.2431   | 1.09313e-05 | True        |     | positive  | 10.33*** [3.65, 29.2]  |
| Language sensitivity | Gate1 | Exclude dominant language | LanguagePR 100   | 100 | Log_PR_SIR   | OR     | 1.28118     | 0.985955   | 1.66479   | 0.063721    | False       |     | positive  | 1.28 [0.99, 1.66]      |
| Language sensitivity | Gate1 | Exclude top 2 languages   | LanguagePR 70    | 70  | Context      | OR     | 3.39731     | 0.997674   | 11.5686   | 0.050436    | False       |     | positive  | 3.40 [1.00, 11.57]     |
| Language sensitivity | Gate1 | Exclude top 2 languages   | LanguagePR 70    | 70  | Specificity  | OR     | 0.580421    | 0.135241   | 2.49103   | 0.464202    | False       |     | negative  | 0.58 [0.14, 2.49]      |
| Language sensitivity | Gate1 | Exclude top 2 languages   | LanguagePR 70    | 70  | Verification | OR     | 10.87       | 3.28817    | 35.9336   | 9.18473e-05 | True        |     | positive  | 10.87*** [3.29, 35.9]  |
| Language sensitivity | Gate1 | Exclude top 2 languages   | LanguagePR 70    | 70  | Log_PR_SIR   | OR     | 1.2232      | 0.899815   | 1.66281   | 0.198412    | False       |     | positive  | 1.22 [0.90, 1.66]      |
| Language sensitivity | Gate2 | Exclude dominant language | LanguagePR 50    | 50  | Context      | AME    | 0.127515    | -0.182533  | 0.437562  | 0.420194    | False       |     | positive  | 0.128 [-0.183, 0.43]   |
| Language sensitivity | Gate2 | Exclude dominant language | LanguagePR 50    | 50  | Specificity  | AME    | 0.0615353   | 0.23889    | 0.36196   | 0.688086    | False       |     | positive  | 0.062 [-0.239, 0.36]   |
| Language sensitivity | Gate2 | Exclude dominant language | LanguagePR 50    | 50  | Verification | AME    | -0.0579782  | 0.321507   | 0.205551  | 0.666319    | False       |     | negative  | -0.058 [-0.322, 0.206] |
| Language sensitivity | Gate2 | Exclude dominant language | LanguagePR 50    | 50  | Log_PR_SIR   | AME    | -0.0323658  | 0.106974   | 0.0422426 | 0.395186    | False       |     | negative  | -0.032 [-0.107, 0.043] |
| Language sensitivity | Gate2 | Exclude top 2 languages   | LanguagePR 40    | 40  | Context      | AME    | 0.0841155   | 0.264401   | 0.432632  | 0.636182    | False       |     | positive  | 0.084 [-0.264, 0.43]   |
| Language sensitivity | Gate2 | Exclude top 2 languages   | LanguagePR 40    | 40  | Specificity  | AME    | 0.0339817   | 0.308107   | 0.37607   | 0.845632    | False       |     | positive  | 0.034 [-0.308, 0.37]   |
| Language sensitivity | Gate2 | Exclude top 2 languages   | LanguagePR 40    | 40  | Verification | AME    | -0.0823957  | 0.381954   | 0.217163  | 0.589816    | False       |     | negative  | -0.082 [-0.382, 0.217] |
| Language sensitivity | Gate2 | Exclude top 2 languages   | LanguagePR 40    | 40  | Log_PR_SIR   | AME    | -0.0150568  | 0.100125   | 0.070011  | 0.72866     | False       |     | negative  | -0.015 [-0.100, 0.07]  |
| PR-size outlier      | Gate1 | Exclusivity               | PRSizeV + PR 120 | 120 | Context      | OR     | 2.08292     | 0.89794    | 4.83169   | 0.087412    | False       |     | positive  | 2.08 [0.90, 4.83]      |
| PR-size outlier      | Gate1 | Exclusivity               | PRSizeV + PR 120 | 120 | Specificity  | OR     | 0.93567     | 0.331314   | 2.64244   | 0.900106    | False       |     | negative  | 0.94 [0.33, 2.64]      |
| PR-size outlier      | Gate1 | Exclusivity               | PRSizeV + PR 120 | 120 | Verification | OR     | 7.83367     | 3.22196    | 19.0463   | 5.59773e-06 | True        |     | positive  | 7.83*** [3.22, 19.0]   |
| PR-size outlier      | Gate1 | Exclusivity               | PRSizeV + PR 120 | 120 | Log_PR_SIR   | OR     | 1.40335     | 1.10236    | 1.78652   | 0.005938    | True        |     | positive  | 1.40** [1.10, 1.79]    |
| PR-size outlier      | Gate1 | Exclusivity               | PRSizeV + PR 120 | 120 | Context      | OR     | 2.25391     | 0.958594   | 5.29956   | 0.062459    | False       |     | positive  | 2.25 [0.96, 5.30]      |
| PR-size outlier      | Gate1 | Exclusivity               | PRSizeV + PR 120 | 120 | Specificity  | OR     | 0.913981    | 0.323584   | 2.58159   | 0.865185    | False       |     | negative  | 0.91 [0.32, 2.58]      |
| PR-size outlier      | Gate1 | Exclusivity               | PRSizeV + PR 120 | 120 | Verification | OR     | 7.8077      | 3.1985     | 19.059    | 6.37785e-06 | True        |     | positive  | 7.81*** [3.20, 19.0]   |
| PR-size outlier      | Gate1 | Exclusivity               | PRSizeV + PR 120 | 120 | Log_PR_SIR   | OR     | 1.38651     | 1.07654    | 1.78573   | 0.011367    | True        |     | positive  | 1.39* [1.08, 1.79]     |
| PR-size outlier      | Gate2 | Exclusivity               | PRSizeV + PR 80  | 80  | Context      | AME    | 0.133179    | -0.101868  | 0.368226  | 0.266774    | False       |     | positive  | 0.133 [-0.102, 0.36]   |
| PR-size outlier      | Gate2 | Exclusivity               | PRSizeV + PR 80  | 80  | Specificity  | AME    | -0.0127385  | 0.244567   | 0.21909   | 0.914237    | False       |     | negative  | -0.013 [-0.245, 0.219] |
| PR-size outlier      | Gate2 | Exclusivity               | PRSizeV + PR 80  | 80  | Verification | AME    | -0.00925981 | 0.121939   | 0.200871  | 0.931173    | False       |     | negative  | -0.009 [-0.219, 0.201] |
| PR-size outlier      | Gate2 | Exclusivity               | PRSizeV + PR 80  | 80  | Log_PR_SIR   | AME    | -0.038691   | 0.101189   | 0.0238074 | 0.224993    | False       |     | negative  | -0.039 [-0.101, 0.024] |
| PR-size outlier      | Gate2 | Exclusivity               | PRSizeV + PR 80  | 80  | Context      | AME    | 0.147784    | -0.0956301 | 0.391199  | 0.234065    | False       |     | positive  | 0.148 [-0.096, 0.39]   |
| PR-size outlier      | Gate2 | Exclusivity               | PRSizeV + PR 80  | 80  | Specificity  | AME    | -0.0127468  | 0.248322   | 0.222829  | 0.915544    | False       |     | negative  | -0.013 [-0.248, 0.223] |

| Family             | Gate   | Scenario                               | Specification                  | N    | Term         | Metric      | Estimate   | CI_Low     | CI_High   | p_value     | Significant | CI_Direction | Formatted                |
|--------------------|--------|----------------------------------------|--------------------------------|------|--------------|-------------|------------|------------|-----------|-------------|-------------|--------------|--------------------------|
| PR-size outlier    | Gate 2 | Exclude top 5% PR-size                 | Size + PR-size                 | 842  | Verification | AME         | -0.000971  | -0.223174  | 0.22123   | 0.99316     | False       | negative     | -0.001 [-0.223, 0.221]   |
| PR-size outlier    | Gate 2 | Exclude top 5% PR-size                 | Size + PR-size                 | 842  | Log_PR_size  | AME         | -0.0402186 | -0.109481  | 0.0290438 | 0.255082    | False       | negative     | -0.040 [-0.109, 0.029]   |
| PR-size outlier    | Access | Exclude top 1% PR-size                 | Size + PR-size                 | 2581 | Context      | HR          | 1.02903    | 0.805731   | 1.131421  | 0.818656    | False       | positive     | 1.03 [0.81, 1.31]        |
| PR-size outlier    | Access | Exclude top 1% PR-size                 | Size + PR-size                 | 2581 | Specificity  | HR          | 1.06444    | 0.810009   | 1.39878   | 0.654102    | False       | positive     | 1.06 [0.81, 1.40]        |
| PR-size outlier    | Access | Exclude top 1% PR-size                 | Size + PR-size                 | 2581 | Verification | HR          | 0.847656   | 0.651359   | 1.10311   | 0.218766    | False       | negative     | 0.85 [0.65, 1.10]        |
| PR-size outlier    | Access | Exclude top 1% PR-size                 | Size + PR-size                 | 2581 | Log_PR_size  | HR          | 0.937227   | 0.875951   | 1.00279   | 0.0602108   | False       | negative     | 0.94 [0.88, 1.00]        |
| PR-size outlier    | Access | Exclude top 1% PR-size                 | Size + PR-size                 | 2581 | Context      | HR          | 1.62168    | 0.885729   | 2.96913   | 0.117171    | False       | positive     | 1.62 [0.89, 2.97]        |
| PR-size outlier    | Access | Exclude top 1% PR-size                 | Size + PR-size                 | 2581 | Specificity  | HR          | 0.633455   | 0.341034   | 1.17662   | 0.148406    | False       | negative     | 0.63 [0.34, 1.18]        |
| PR-size outlier    | Access | Exclude top 1% PR-size                 | Size + PR-size                 | 2581 | Verification | HR          | 1.70702    | 0.992656   | 2.93547   | 0.053193    | False       | positive     | 1.71 [0.99, 2.94]        |
| PR-size outlier    | Access | Exclude top 1% PR-size                 | Size + PR-size                 | 2581 | Log_PR_size  | HR          | 0.774604   | 0.669356   | 0.8964    | 0.000608321 | True        | negative     | 0.77*** [0.67, 0.90]     |
| PR-size outlier    | Access | Exclude top 5% PR-size                 | Size + PR-size                 | 2381 | Context      | HR          | 1.03575    | 0.806987   | 1.32936   | 0.782657    | False       | positive     | 1.04 [0.81, 1.33]        |
| PR-size outlier    | Access | Exclude top 5% PR-size                 | Size + PR-size                 | 2381 | Specificity  | HR          | 1.08197    | 0.816667   | 1.43346   | 0.583066    | False       | positive     | 1.08 [0.82, 1.43]        |
| PR-size outlier    | Access | Exclude top 5% PR-size                 | Size + PR-size                 | 2381 | Verification | HR          | 0.815483   | 0.620227   | 1.07221   | 0.144095    | False       | negative     | 0.82 [0.62, 1.07]        |
| PR-size outlier    | Access | Exclude top 5% PR-size                 | Size + PR-size                 | 2381 | Log_PR_size  | HR          | 0.953612   | 0.884808   | 1.02777   | 0.213794    | False       | negative     | 0.95 [0.88, 1.03]        |
| PR-size outlier    | Access | Exclude top 5% PR-size                 | Size + PR-size                 | 2381 | Context      | HR          | 1.56533    | 0.835372   | 2.93314   | 0.161941    | False       | positive     | 1.57 [0.84, 2.93]        |
| PR-size outlier    | Access | Exclude top 5% PR-size                 | Size + PR-size                 | 2381 | Specificity  | HR          | 0.632314   | 0.332598   | 1.20211   | 0.161992    | False       | negative     | 0.63 [0.33, 1.20]        |
| PR-size outlier    | Access | Exclude top 5% PR-size                 | Size + PR-size                 | 2381 | Verification | HR          | 1.733      | 0.996276   | 3.01453   | 0.0515598   | False       | positive     | 1.73 [1.00, 3.01]        |
| PR-size outlier    | Access | Exclude top 5% PR-size                 | Size + PR-size                 | 2381 | Log_PR_size  | HR          | 0.731339   | 0.625027   | 0.855734  | 9.4638e-05  | True        | negative     | 0.73*** [0.63, 0.86]     |
| Gate 2 alternative | Gate 2 | Resizetories                           | Ordinal logistic               | 38   | Context      | Coefficient | 0.969038   | 0.078135   | 1.85994   | 0.0330184   | True        | positive     | 0.97* [0.08, 1.86]       |
| Gate 2 alternative | Gate 2 | Resizetories                           | Ordinal logistic               | 38   | Specificity  | Coefficient | 0.16346    | -0.796192  | 1.12311   | 0.738497    | False       | positive     | 0.16 [-0.80, 1.12]       |
| Gate 2 alternative | Gate 2 | Resizetories                           | Ordinal logistic               | 38   | Verification | Coefficient | 0.394819   | 1.21647    | 0.426829  | 0.346293    | False       | negative     | -0.39 [-1.22, 0.43]      |
| Gate 2 alternative | Gate 2 | Resizetories                           | Ordinal logistic               | 38   | Log_PR_size  | Coefficient | 0.214976   | -0.460195  | 0.030244  | 0.0857547   | False       | negative     | -0.21 [-0.46, 0.03]      |
| Gate2 binary       | Gate 2 | Low vs low reuse                       | Binary logistic                | 160  | Context      | OR          | 2.91747    | 0.84306    | 10.0961   | 0.0909447   | False       | positive     | 2.92 [0.84, 10.10]       |
| Gate2 binary       | Gate 2 | Low vs low reuse                       | Binary logistic                | 160  | Specificity  | OR          | 1.23437    | 0.345707   | 4.40739   | 0.745744    | False       | positive     | 1.23 [0.35, 4.41]        |
| Gate2 binary       | Gate 2 | Low vs low reuse                       | Binary logistic                | 160  | Verification | OR          | 0.764335   | 0.218298   | 2.67619   | 0.674242    | False       | negative     | 0.76 [0.22, 2.68]        |
| Gate2 binary       | Gate 2 | Low vs low reuse                       | Binary logistic                | 160  | Log_PR_size  | OR          | 0.760425   | 0.516874   | 1.11874   | 0.164417    | False       | negative     | 0.76 [0.52, 1.12]        |
| Aggregate PQS      | Gate 0 | PQS only                               | PQS + PR-size                  | 218  | PQS          | OR          | 2.94439    | 2.1454     | 4.04093   | 2.29569e-11 | True        | positive     | 2.94*** [2.15, 4.04]     |
| Aggregate PQS      | Gate 0 | PQS only                               | PQS + PR-size                  | 218  | Log_PR_size  | OR          | 1.05018    | 0.877445   | 1.25691   | 0.593356    | False       | positive     | 1.05 [0.88, 1.26]        |
| Aggregate PQS      | Gate 0 | PQS only                               | PQS + PR-size                  | 1241 | PQS          | OR          | 2.70132    | 1.78583    | 4.08611   | 2.52339e-06 | True        | positive     | 2.70*** [1.79, 4.09]     |
| Aggregate PQS      | Gate 0 | PQS only                               | PQS + PR-size                  | 1241 | Log_PR_size  | OR          | 1.39301    | 1.12023    | 1.73221   | 0.00287251  | True        | positive     | 1.39** [1.12, 1.73]      |
| Aggregate PQS      | Gate 0 | PQS only                               | PQS + PR-size                  | 882  | PQS          | AME         | 0.0171407  | 0.0778767  | 0.112158  | 0.723662    | False       | positive     | 0.017 [-0.078, 0.112]    |
| Aggregate PQS      | Gate 0 | PQS only                               | PQS + PR-size                  | 882  | Log_PR_size  | AME         | -0.0379279 | -0.0974745 | 0.0216186 | 0.211888    | False       | negative     | -0.038 [-0.097, 0.022]   |
| Prompt-length      | Gate 0 | Add log(prompt_tokens) + prompt length | Size + PR-size + prompt length | 218  | Context      | DR          | 2.16152    | 1.13641    | 4.11134   | 0.0187837   | True        | positive     | 2.16* [1.14, 4.11]       |
| Prompt-length      | Gate 0 | Add log(prompt_tokens) + prompt length | Size + PR-size + prompt length | 218  | Specificity  | DR          | 60.3944    | 8.10811    | 449.857   | 6.26159e-05 | True        | positive     | 60.39*** [8.11, 449.857] |
| Prompt-length      | Gate 0 | Add log(prompt_tokens) + prompt length | Size + PR-size + prompt length | 218  | Verification | DR          | 0.834094   | 0.389925   | 1.78422   | 0.640073    | False       | negative     | 0.83 [0.39, 1.78]        |
| Prompt-length      | Gate 0 | Add log(prompt_tokens) + prompt length | Size + PR-size + prompt length | 218  | Log_PR_size  | DR          | 1.11227    | 0.911245   | 1.35765   | 0.295489    | False       | positive     | 1.11 [0.91, 1.36]        |
| Prompt-length      | Gate 0 | Add log(prompt_tokens) + prompt length | Size + PR-size + prompt length | 218  | Log_PR_size  | Tokens      | 2.55836    | 0.30117    | 21.7325   | 0.389481    | False       | positive     | 2.56 [0.30, 21.73]       |
| Prompt-length      | Gate 0 | Add log(prompt_tokens) + prompt length | Size + PR-size + prompt length | 1241 | Context      | DR          | 2.14048    | 0.909116   | 5.03968   | 0.0815297   | False       | positive     | 2.14 [0.91, 5.04]        |

| Family             | Gate    | Scenario                                           | Specification            | N   | Term              | Metric      | Estimate    | CI_Low     | CI_High   | p_value     | Significant | CI_Direction | Formatted              |
|--------------------|---------|----------------------------------------------------|--------------------------|-----|-------------------|-------------|-------------|------------|-----------|-------------|-------------|--------------|------------------------|
| Prompt-length      | Control | Add log(prompt_tokens/V + PR_size + prompt_length) | Size + prompt length     | 171 | Spreading         | OR          | 0.556016    | 0.194684   | 1.58798   | 0.272975    | False       | negative     | 0.56 [0.19, 1.59]      |
| Prompt-length      | Control | Add log(prompt_tokens/V + PR_size + prompt_length) | Size + prompt length     | 171 | Verification      | OR          | 6.9634      | 2.81381    | 17.2325   | 2.69666e-05 | True        | positive     | 6.96*** [2.81, 17.23]  |
| Prompt-length      | Control | Add log(prompt_tokens/V + PR_size + prompt_length) | Size + prompt length     | 171 | Log PR_Size       | SE          | 1.36285     | 1.07618    | 1.72587   | 0.0101901   | True        | positive     | 1.36* [1.08, 1.73]     |
| Prompt-length      | Control | Add log(prompt_tokens/V + PR_size + prompt_length) | Size + prompt length     | 171 | Log prompt_Tokens | OR          | 18.3295     | 1.82162    | 184.434   | 0.0135461   | True        | positive     | 18.33* [1.82, 184.43]  |
| Prompt-length      | Control | Add log(prompt_tokens/V + PR_size + prompt_length) | Size + prompt length     | 82  | Context           | AME         | 0.128498    | -0.102812  | 0.359808  | 0.276241    | False       | positive     | 0.128 [-0.103, 0.36]   |
| Prompt-length      | Control | Add log(prompt_tokens/V + PR_size + prompt_length) | Size + prompt length     | 82  | Spreading         | AME         | -0.00493589 | -0.0247281 | 0.237409  | 0.968158    | False       | negative     | -0.005 [-0.247, 0.237] |
| Prompt-length      | Control | Add log(prompt_tokens/V + PR_size + prompt_length) | Size + prompt length     | 82  | Verification      | AME         | -0.00741306 | -0.0216899 | 0.202073  | 0.944705    | False       | negative     | -0.007 [-0.217, 0.202] |
| Prompt-length      | Control | Add log(prompt_tokens/V + PR_size + prompt_length) | Size + prompt length     | 82  | Log PR_Size       | SE          | -0.036678   | 0.097841   | 0.0244854 | 0.23986     | False       | negative     | -0.037 [-0.098, 0.024] |
| Prompt-length      | Control | Add log(prompt_tokens/V + PR_size + prompt_length) | Size + prompt length     | 82  | Log prompt_Tokens | AME         | -0.0579665  | -0.541513  | 0.42558   | 0.814243    | False       | negative     | -0.058 [-0.542, 0.426] |
| Hold-out stability | Gate 0  | Stratified 80/20 split                             | Train N=175, 75 Holdout  | 75  | Context           | Coefficient | 0.904959    | nan        | nan       | 0.0149389   | True        | positive     | beta=0.905*            |
| Hold-out stability | Gate 0  | Stratified 80/20 split                             | Train N=175, 75 Holdout  | 75  | Specificity       | Coefficient | 4.03769     | nan        | nan       | 8.81199e-05 | True        | positive     | beta=4.038***          |
| Hold-out stability | Gate 0  | Stratified 80/20 split                             | Train N=175, 75 Holdout  | 75  | Verification      | Coefficient | 0.257766    | nan        | nan       | 0.54301     | False       | negative     | beta=-0.258            |
| Hold-out stability | Gate 0  | Stratified 80/20 split                             | Train N=175, 75 Holdout  | 75  | Log PR_Size       | Coefficient | 0.0478501   | nan        | nan       | 0.702407    | False       | positive     | beta=0.048             |
| Hold-out stability | Gate 1  | Stratified 80/20 split                             | Train N=113, 113 Holdout | 113 | Context           | Coefficient | 0.685885    | nan        | nan       | 0.136527    | False       | positive     | beta=0.686             |
| Hold-out stability | Gate 1  | Stratified 80/20 split                             | Train N=113, 113 Holdout | 113 | Specificity       | Coefficient | 0.0220572   | nan        | nan       | 0.945385    | False       | negative     | beta=-0.039            |
| Hold-out stability | Gate 1  | Stratified 80/20 split                             | Train N=113, 113 Holdout | 113 | Verification      | Coefficient | 2.06749     | nan        | nan       | 2.42577e-05 | True        | positive     | beta=2.067***          |
| Hold-out stability | Gate 1  | Stratified 80/20 split                             | Train N=113, 113 Holdout | 113 | Log PR_Size       | Coefficient | 0.209699    | nan        | nan       | 0.0914981   | False       | positive     | beta=0.210             |
| Hold-out stability | Gate 2  | Stratified 80/20 split                             | Train N=71, 71 Holdout   | 71  | Context           | Coefficient | 0.438573    | nan        | nan       | 0.479228    | False       | positive     | beta=0.439             |
| Hold-out stability | Gate 2  | Stratified 80/20 split                             | Train N=71, 71 Holdout   | 71  | Specificity       | Coefficient | 0.15706     | nan        | nan       | 0.800494    | False       | negative     | beta=-0.157            |
| Hold-out stability | Gate 2  | Stratified 80/20 split                             | Train N=71, 71 Holdout   | 71  | Verification      | Coefficient | 0.0543526   | nan        | nan       | 0.923554    | False       | positive     | beta=0.054             |
| Hold-out stability | Gate 2  | Stratified 80/20 split                             | Train N=71, 71 Holdout   | 71  | Log PR_Size       | Coefficient | 0.188858    | nan        | nan       | 0.239433    | False       | negative     | beta=-0.189            |